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Mask-Aware Policy Gradients

UT Austin Corrects a Fundamental Bias in RL Training for Masked Diffusion Language Models by Properly Accounting for Both Token and Masking Decisions

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Masked Diffusion Language Models (MDLMs) generate text by iteratively unmasking tokens over multiple steps. Applying reinforcement learning to MDLMs — to improve performance on math reasoning, code generation, or instruction following — requires a valid policy gradient. But all existing RL approaches for MDLMs compute a policy gradient that ignores one of the model’s two fundamental decisions: not just which tokens to predict, but which positions to remask at each step. TRACE-MDLM corrects this omission with a two-stage action MDP formulation, raising GSM8K accuracy from 79.3% to 87.1% and MBPP from 41.2% to 53.4%.

AT A GLANCE

Problem: RL for MDLMs ignores the masking schedule decision in the policy gradient, producing a biased update that misattributes credit to token choices.

Why it matters: MDLMs are an increasingly competitive alternative to autoregressive LMs; reliable RL is necessary for closing the remaining gap to AR models on reasoning tasks.

Method: Two-stage action MDP: the policy gradient decomposes into a token term and a masking term; a lightweight MLP head is trained to parameterize the masking policy.

Result: 87.1% on GSM8K (+7.8 pp over token-only baseline); 53.4% on MBPP (+12.2 pp); 40% lower gradient variance.

Evidence: Evaluated on GSM8K and MBPP; ablations on 3B-scale MDLM; accepted at COLM 2026.

The Big Picture

Masked Diffusion Language Models are a promising alternative to autoregressive (AR) LMs: they are intrinsically parallelizable across token positions, support flexible bidirectional context, and can be sampled at adjustable quality-speed tradeoffs by varying the number of denoising steps. MDLMs have closed much of the gap to AR

models on language modeling benchmarks, but lag behind on structured reasoning tasks like mathematics and code.

Reinforcement learning from verifiable rewards — the technique that drove major gains for AR reasoning models via GRPO, PPO, and similar methods — is the natural next step for MDLMs. The obstacle is that naive application of AR policy gradient methods to MDLMs produces a biased estimator, because it ignores a decision that is structurally part of the MDLM generation process: the masking schedule.

Why Existing Approaches Fall Short

Standard RL for AR models computes the log-probability of each generated token and uses these log-probabilities directly in the policy gradient. For MDLMs, the analogous computation scores each predicted token at each unmasked position — ignoring that the model also decided which positions to keep unmasked (and which to remask) at each step.

The masking schedule is not a fixed algorithmic artifact: it is a learned policy decision. A given sequence of T denoising steps involves T masking decisions, each dependent on the model’s output and the task. Ignoring these decisions is equivalent to computing the policy gradient for an AR model while ignoring every other token in the generation — the gradient is biased and the resulting training signal miscredits token choices for outcomes that were actually driven by masking decisions.

Prior approaches including SPG and GRPO-for-MDLMs apply variance reduction techniques to the token gradient without addressing this structural bias.

Core Mechanism

The paper formalizes MDLM generation as a two-stage MDP per denoising step. At step t , the model observes a partially masked sequence $\mathbf{x}_t$ and takes two sequential actions:

1. **Token action** a_t^{tok} : For each masked position $i \in \mathcal{M}_t$, predict the unmasked token x_i from the MDLM network. This recovers a fully unmasked intermediate sequence $\hat{\mathbf{x}}_t$.
2. **Masking action** a_t^{mask} : Select a subset of positions in $\hat{\mathbf{x}}_t$ to remask, producing $\mathbf{x}_{t+1}$ for the next step. This is parameterized by a lightweight MLP over the MDLM hidden states.

The full policy gradient decomposes as:

$$\nabla_{\theta} J = \mathbb{E}_{\pi} \left[\hat{A}_t (\nabla_{\theta} \log \pi_{\theta}(a_t^{\text{tok}} \mid \mathbf{x}_t) + \nabla_{\theta} \log \pi_{\theta}(a_t^{\text{mask}} \mid \hat{\mathbf{x}}_t)) \right]$$

where $\hat{A}_t$ is the advantage estimate at step t .

Technical Formulation

The masking policy $\pi_\theta(a_t^{\text{mask}} \mid \hat{\mathbf{x}}_t)$ is a factored Bernoulli over positions:

$$\pi_\theta(a_t^{\text{mask}} \mid \hat{\mathbf{x}}_t) = \prod_{i=1}^L \text{Bernoulli}(m_i; \sigma(\mathbf{W}\mathbf{h}_i + b))$$

where $\mathbf{h}_i$ is the MDLM hidden state at position i after the token action, $\mathbf{W} \in \mathbb{R}^{1 \times d}$ and $b \in \mathbb{R}$ are the MLP parameters, and $m_i \in \{0, 1\}$ is the masking decision. The MLP is trained jointly with the denoising network.

The advantage $\hat{A}_t$ is estimated using Generalized Advantage Estimation (GAE) with $\lambda = 0.95$ and a value function head trained on the MDLM encoder representations.

Experimental Findings

Method	GSM8K	MBPP
MDLM-3B (no RL)	52.1%	28.7%
Token-only REINFORCE	79.3%	41.2%
SPG	81.4%	44.7%
Mask-Aware PG (ours)	87.1%	53.4%

The masking term alone contributes approximately 3–4 percentage points; jointly optimizing both token and masking gradients provides the full 7–12 percentage point improvement over token-only methods.

Gradient norm variance is 40% lower with Mask-Aware PG, explaining the smoother training curves and reduced need for gradient clipping.

Ablations and Interpretation

Token gradient only: Removing the masking term reproduces the SPG/REINFORCE baseline, confirming the masking term as the source of improvement.

Masking term only: Training with only the masking gradient (no token gradient) fails to converge, confirming both terms are jointly necessary.

Masking schedule analysis: Models trained with Mask-Aware PG learn to unmask tokens in a semantically coherent order: in math problems, nouns and verbs are unmasked first, modifiers last. This learned curriculum differs qualitatively from the uniform-random schedule used at initialization.

Scaling: The advantage grows with problem difficulty. For easy single-step GSM8K problems the gap is <3 percentage points; for multi-step problems requiring 4+ intermediate computations the gap is 8+ percentage points, consistent with the masking schedule being most important when sequential reasoning structure matters.

Reference

Haran Raajesh, Kulin Shah, Adam Klivans, Philipp Krähenbühl. **Mask-Aware Policy Gradients for Diffusion Language Models.** COLM 2026 / arXiv:2607.15200, July 2026. <https://arxiv.org/abs/2607.15200>